Question # 1

Write the following expression in the form of $a+ib$.

$\frac{3+i}{2-i}$, $\frac{4+10i}{-9+7i}$

Answer:

$$\frac{3+i}{2-i}=\frac{3+i}{2-i}*\frac{2+i}{2+i}=\frac{6+3i+2i+i^2}{4+2i-2i-i^2}=\frac{6+5i+(-1)}{4-(-1)}=\frac{5+5i}{5}$$

$$\frac{4+10i}{-9+7i}*\frac{-9-7i}{-9-7i}=\frac{-36-28i-90i-70i^2}{81+63i-63i-49i^2}=\frac{-36-118i+70}{81+49}=\frac{34-118i}{130}$$

Question # 2

Write in set builder notation

1. A= Set of first 100 odd numbers
2. A= Set of Natural numbers starting from 2 to 40.
3. A= Set of all even numbers greater than 50.

Answer:

A = {x| x is set of first 100 odd numbers}

A = {x|2≤x≤40 and x is set of natural numbers}

A = {x|x>50 and x is set of even numbers}

Question # 3

List the elements of the set (roster method)

1. $A=\{x \mid x\in E^{+} \wedge x\leq 10\}$
2. $A=\{x \mid x^2=9,\ x\ is\ even\}$
3. $A=\{x \mid x\in N\ \wedge 2\leq x\leq 10\}$

Answer:

A = {0,2,4,6,8,10}

A = ∅

A = {2,3,4,5,6,7,8,9,10}

Question # 4

Let A be the set of all even numbers.

Let B be the set of all Number less than 100.

Is A a subset of B? or

Is B a subset of A?

Answer:

$A \not\subset B$

$B \not\subset A$

List all elements in these sets and verify the result.

Question # 5

The complex numbers Z is given by

$$Z = \frac{i-1}{i}$$

Find modulus of Z .

Answer:

$$Z = \frac{i-1}{i}$$

$$\frac{i-1}{i} * \frac{i}{i} = \frac{i^2 - i}{i^2} = \frac{-(1-i)}{-1} = \frac{1+i}{1}$$

$$|z| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

Question # 6

Evaluate

1. $(2+i)(3+i)$
2. $(4+5i)(-2-2i)$
3. $(-1-i)(-2-2i)$
4. i^3

Solution

Solution:

$$(2+i)(3+i) = 6+2i+3i+i^2 = 6+5i(-1) = 5+5i$$

$$(4+5i)(-2-2i) = -8-8i-10i-10i^2 = -8-18i-10(-1) = -8-18i+10 = 2-18i$$

$$(-1-i)(-2-2i) = 2+2i+2i+2i^2 = 2+4i+2(-1) = 4i$$

$$i^3 = i*i*i = (i)^2(i) = (-1)(\sqrt{-1}) = -i$$

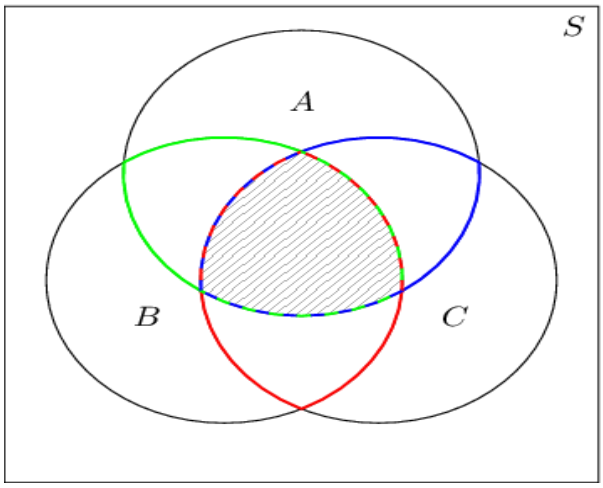
Question # 7

The Venn diagram given below shows the sets A, B, and C. Shade the following sets

i. $A \cup (B \cap C)$

ii. $A \cup (B \cap C)^c$

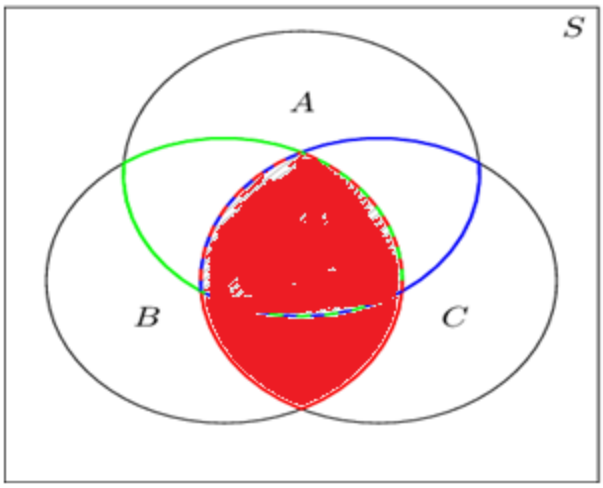
iii. $A \cap (B \cup C)$



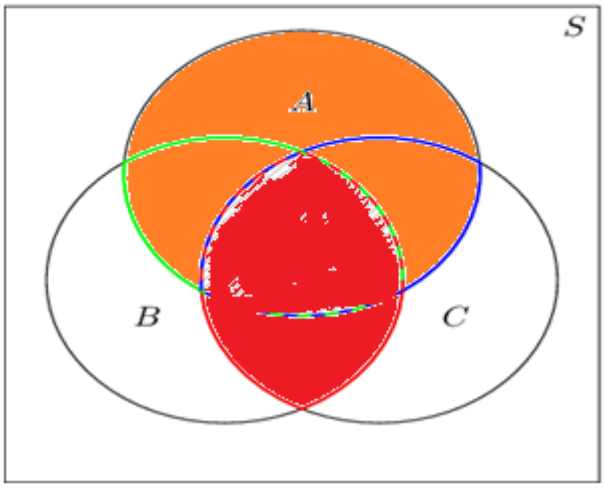
Solution

Solution:

i. $A \cup (B \cap C)$



B intersection C is shown in red. If we take the union of this red portion with the A then it will be like



Question # 8

What is the cardinality of these sets?

A = Set of even integers

B = Set of prime numbers less than 30

Solution

Cardinality of A is infinity

Cardinality of B is 10

Question # 9

i) $z = -7 + 5i$

ii) $z = 3i$

Find $z + z^*$, $z.z^*$, $z - z^*$

Solution

$$z = -7 + 5i$$

$$z^* = -7 - 5i$$

$$z + z^* = -7 + 5i + (-7 - 5i)$$

$$= -7 + 5i - 7 - 5i = -14$$

Similarly do others

Question # 10

Evaluate

i. $\left|\frac{2+2i}{-5+3i}\right|$

ii. $\left|\frac{1+i}{2+3i}\right|$

Solution

See the following related example.

$\left|\frac{4+7i}{-5+6i}\right|$ =

$$\frac{4+7i}{-5+6i} * \frac{-5-6i}{-5-6i} = \frac{(4+7i)(-5-6\mathrm{i})}{(-5+6i)(-5-6i)}$$

$$= \frac{-20-24i-35i-42i^2}{(-5)^2-(6i)^2}$$

$$= \frac{22-59i}{25+36} = \frac{22-59i}{61} = \frac{22}{61} - \frac{59}{61}i$$

$$\left|\frac{4+7i}{-5+6i}\right| = \sqrt{(\frac{22}{61})^2 + (\frac{59}{61})^2} = \sqrt{\frac{484+3481}{3721}} = \sqrt{\frac{3965}{3721}}$$

Similarly do other parts

Question related to even function

$$f(\mathrm{x}) = x^4 + 5x^2 + 3$$
$$f(-x) = (-x)^4 + 5(-x)^2 + 3$$
$$f(-x) = x^4 + 5x^2 + 3 = f(x)$$

Therefore the given function is even

Similarly do the parts for even functions

Question related to odd

$$f(\mathrm{x}) = x^3 - 8x$$
$$f(-x) = (-x)^3 - 8(-x)$$
$$f(-x) = -x^3 + 8x = -(x^3 - 8x) = -f(x)$$

Therefore the given function is odd

Similarly do questions for odd

Question

Convert to completed squared form

$x^2 + 4x + 6 = x^2 + 4x + 4 - 4 + 6 = (x+2)^2 + 2$

Similarly do questions for converting to completed square form.

See the following examples to solve questions no.14

Example

If the functions *f* and *g* are defined by

$$f(x) = \frac{3}{x}, \qquad 0 < x \leq 3,$$
$$g(x) = 5x - 1, \quad x \in R,$$

then Calculate *gf(2).*

Solution:

gf(x) = g(f(x)) = g(3/x) = 5(3/x) – 1 = (15/x) – 1.

So, *gf(2) = (15/2) – 1 =13/2*

example

If the functions *g* is defined by

$g(x) = 8x - 5, \quad x \in R,$

then find an expression in terms of *x* for $g^{-1}(x)$.

Solution:

To find $g^{-1}(x)$, we write *y = 8x – 5*

→*y + 5 = 8x* →*x = (y + 5)/8.*

Therefore, $g^{-1}(x) = (x + 5)/8$

Example

Let $f(x) = 6x + 1$ and $g(x) = -9x + 5.$ Find the product of f & g *i.e* $(f.g)(x).$

Solution:

$$(f.g)(x) = f(x).g(x) = (6x+1).(-9x+5)$$
$$= -54x^2 + 21x + 5 \ (Ans.)$$

Similarly you can do other questions.